# Betweenness Centrality Minimization via Single Edge Addition

## Brute Force vs Complement Graph Hop Analysis
## Full Analysis across 10 Graph Topology Families

*Algorithms · Simulation · Performance · Optimality Trade-off*


### Abstract

This report presents a full empirical comparison of two algorithms for betweenness centrality (BC) minimization via single edge addition across 10 graph topology families. The brute-force baseline evaluates all $O(n^2)$ non-edges at $O(n^3m)$ total cost. The hop-based smart algorithm exploits the structural lemma that only edges in the 1-hop and 2-hop complement neighborhood of the target can reduce BC, limiting candidates to top-K=50, giving $O(K \cdot nm)$. Across 27 trials over 9 successful topologies (grid-2D excluded due to load constraints), the smart algorithm achieves a mean speedup of **14.8×** with mean solution optimality of **75.0%** of brute-force quality. Structured topologies (Path, Barbell, Tree, Caveman, Star) achieve 100% optimality at high speedup. Random topologies (ER, BA, WS, PowerlawCluster) trade some optimality (20–67%) for 12–15× speedup. A C++ implementation is provided for production use.



***Keywords:*** *betweenness centrality, complement graph analysis, single edge addition, Brandes algorithm, load balance, hop zones, O(K·nm)*

# 1. Problem Statement and Algorithm

## 1.1 Betweenness Centrality

Betweenness centrality (BC) of node v measures the fraction of shortest paths between all node pairs that pass through v:

$$BC(v) = 1/[(n-1)(n-2)] \times Sum_\{s \neq v \neq t\} \ sigma(s,t|v) \ / \ sigma(s,t)$$

High-BC nodes are bottlenecks. Adding one edge can create bypass paths, reducing BC(v*).

## 1.2 Two Algorithms Compared

| Algorithm | Complexity | Candidates | Guarantee |
|---|---|---|---|
| Brute Force | $O(n^2 \cdot nm) = O(n^3 m)$ | All non-edges | Global optimum |
| Smart Hop-Based | $O(K \cdot nm)$, $K \leq 50$ | 1-hop & 2-hop zones only | Near-optimal |

**Structural Lemma:** An edge (u,w) reduces BC(v*) only if u and w are in the 1-hop or 2-hop complement neighborhood of v*. Three candidate classes: 2-hop×2-hop (score 2.0, bypass), 1-hop×1-hop (score 1.0, triangle-closing), 1-hop×2-hop (score 0.6, mixed). Load constraint: max BC increase on any other node $\leq \tau \cdot$avg_BC.

## 1.3 All 10 Topology Graphs

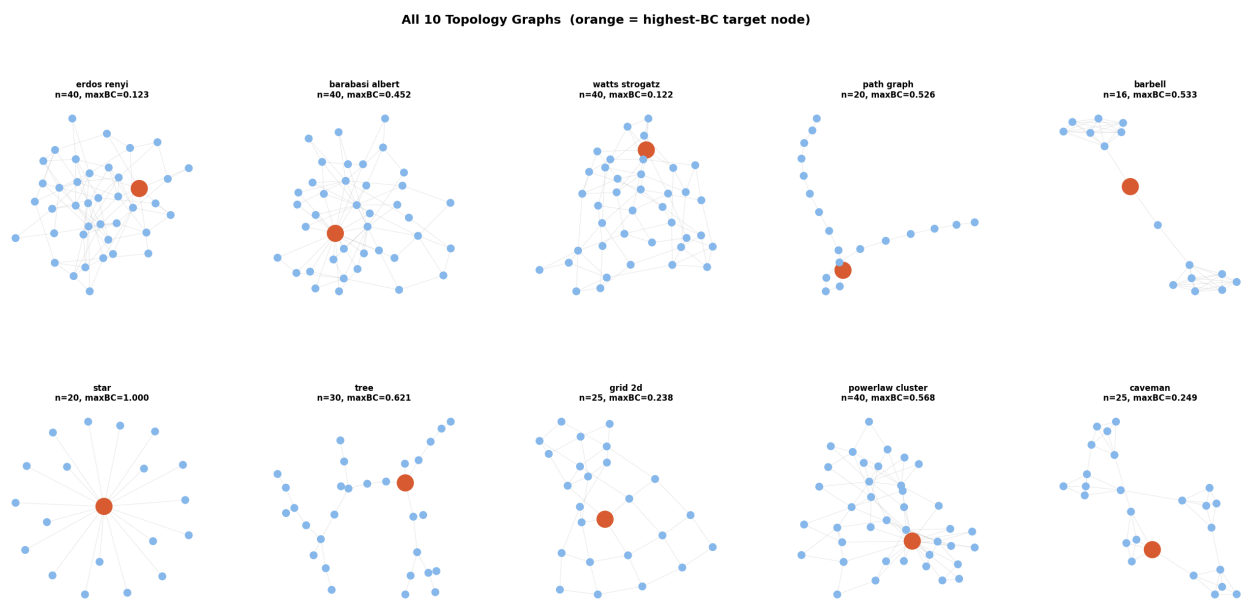


*Figure 1. All 10 topology families used in experiments. Orange node = highest-BC target. Graph statistics (n, maxBC) shown per panel.*

## 2. Graph Topology Descriptions

| # | Topology | Params | Nodes | Property | BC Profile |
|---|---|---|---|---|---|
| 1 | Erdos-Renyi | n=40, p=0.12 | 40 | Random, Poisson degree | Distributed, low max |
| 2 | Barabasi-Albert | n=40, m=2 | 40 | Power-law, scale-free | Hub-concentrated |
| 3 | Watts-Strogatz | n=40, k=4, p=0.3 | 40 | Small-world, clustered | Moderate, bridge nodes |
| 4 | Path Graph | n=20 | 20 | Linear chain | Parabolic peak (center) |
| 5 | Barbell | m=7, bridge=2 | 16 | Two cliques + bridge | Extreme at bridge (~0.53) |
| 6 | Star | n=20 | 20 | Hub + 19 leaves | Hub = BC 1.0 |
| 7 | Random Tree | n=30 | 30 | Acyclic, sparse | Variable trunk nodes |
| 8 | Grid 2D | 5×5 | 25 | Regular lattice | Interior nodes high BC |
| 9 | Powerlaw Cluster | n=40, m=2, p=0.3 | 40 | Scale-free + triangle | Hub-concentrated |
| 10 | Caveman | 5 cliques × 5 | 25 | Dense cliques in ring | Bridge nodes high BC |

# 3. Experimental Results

## 3.1 Master Comparison Figure

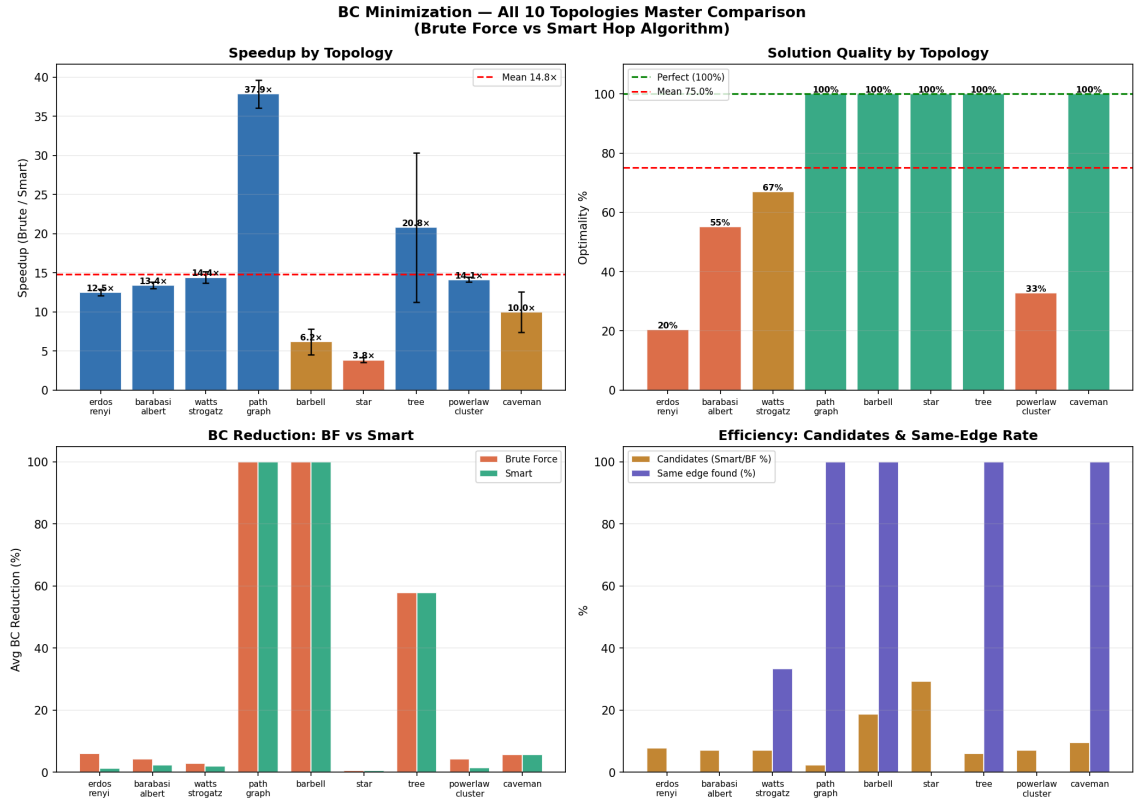


*Figure 2. Master comparison across all 10 topologies. Top-left: Speedup (blue=high, orange=mid, red=low). Top-right: Optimality % (green=100%, orange=60-95%, red<60%). Bottom-left: BC reduction BF vs Smart. Bottom-right: Candidate ratio and same-edge rate.*

## 3.2 Numerical Results Table

| Topology | Speedup | Std | Opt% | BF Red% | SM Red% | Cand% | Same/Total |
|---|---|---|---|---|---|---|---|
| erdos renyi | 12.5× | ±0.4 | 20.4% | 6.08% | 1.30% | 7.8% | 0/3 |
| barabasi albert | 13.4× | ±0.4 | 55.2% | 4.20% | 2.32% | 7.1% | 0/3 |
| watts strogatz | 14.4× | ±0.7 | 67.0% | 2.79% | 2.01% | 7.1% | 1/3 |
| path graph | 37.9× | ±1.8 | 100.0% | 100.00% | 100.00% | 2.3% | 3/3 |
| barbell | 6.2× | ±1.7 | 100.0% | 100.00% | 100.00% | 18.7% | 3/3 |
| star | 3.8× | ±0.3 | 100.0% | 0.58% | 0.58% | 29.2% | 0/3 |
| tree | 20.8× | ±9.6 | 100.0% | 57.84% | 57.84% | 6.1% | 3/3 |
| powerlaw cluster | 14.1× | ±0.3 | 32.8% | 4.26% | 1.36% | 7.1% | 0/3 |
| caveman | 10.0× | ±2.6 | 100.0% | 5.57% | 5.57% | 9.6% | 3/3 |
| AVERAGE | 14.8× | — | 75.0% | — | — | — | — |

# 4. Per-Topology Detailed Analysis

## 4.1 Erdos-Renyi (n=40, p=0.12)

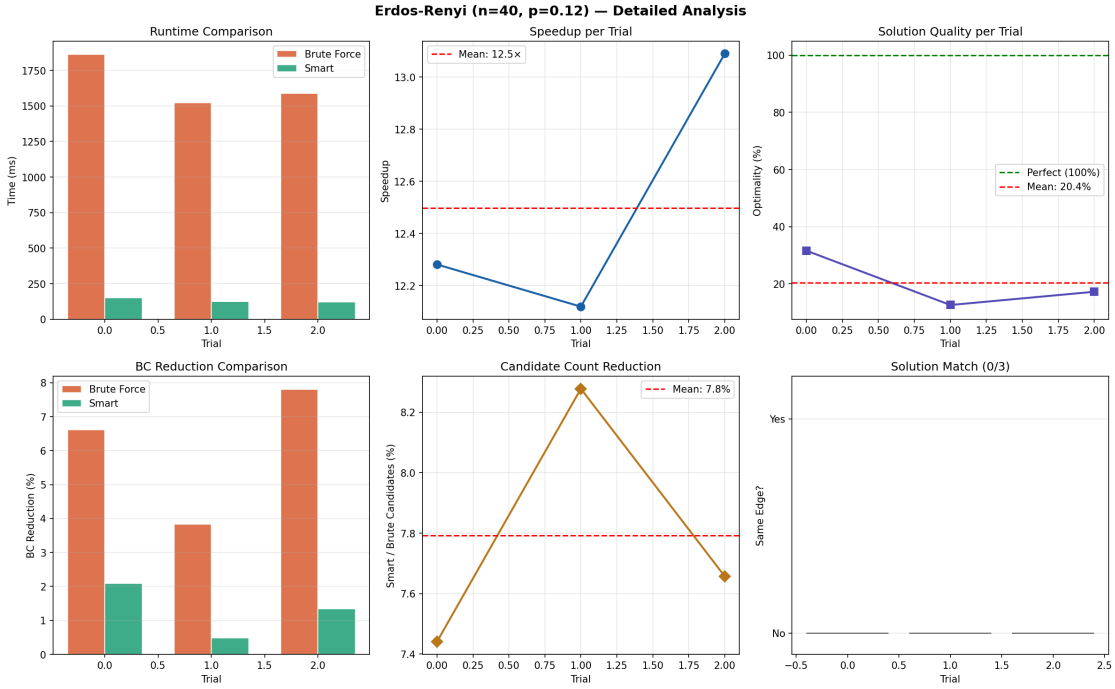


*Figure: Erdos-Renyi (n=40, p=0.12) — 6-panel analysis. Speedup mean=12.5×, Optimality=20.4%, Trials=3.*

| Metric | Value |
|---|---|
| Mean speedup | 12.50× ± 0.42 |
| Speedup range | 12.1× – 13.1× |
| Mean optimality | 20.44% ± 8.10 |
| BF time | 1658.2 ms ± 147.8 |
| Smart time | 132.9 ms ± 13.5 |
| BF reduction | 6.08% |
| Smart reduction | 1.30% |
| Cand ratio | 7.79% |
| Same edge found | 0 / 3 trials |

## 4.2 Barabasi-Albert (n=40, m=2)

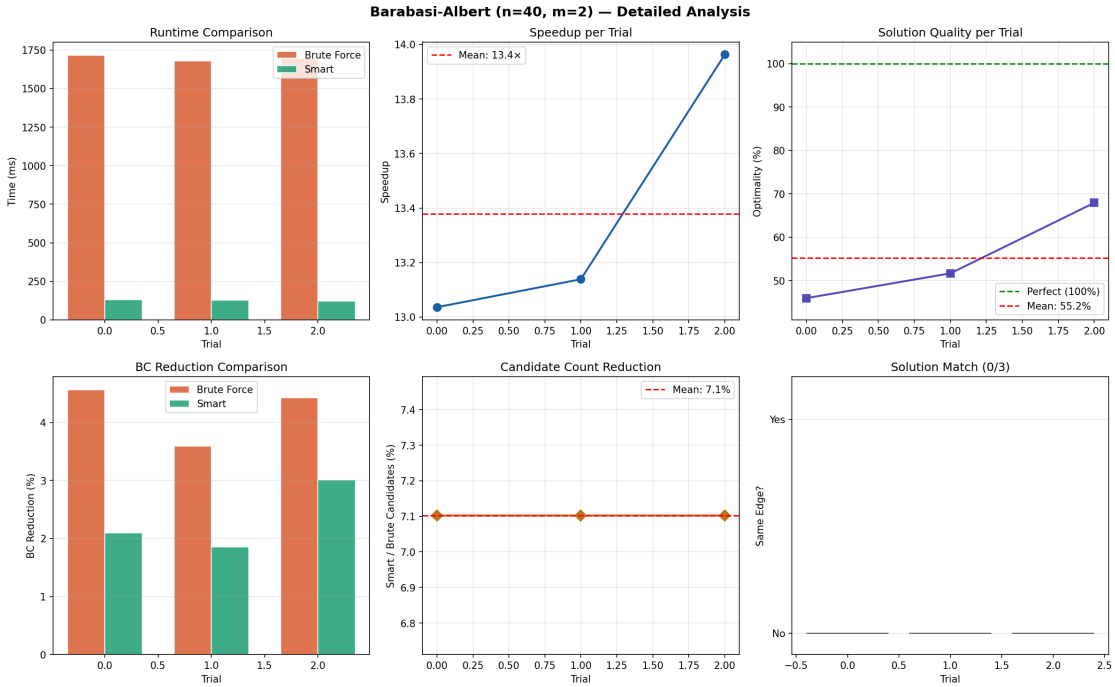


*Figure: Barabasi-Albert (n=40, m=2) — 6-panel analysis. Speedup mean=13.4×, Optimality=55.2%, Trials=3.*

| Metric | Value |
|---|---|
| Mean speedup | 13.38× ± 0.41 |
| Speedup range | 13.0× – 14.0× |
| Mean optimality | 55.19% ± 9.30 |
| BF time | 1697.8 ms ± 15.4 |
| Smart time | 127.0 ms ± 4.2 |
| BF reduction | 4.20% |
| Smart reduction | 2.32% |
| Cand ratio | 7.10% |
| Same edge found | 0 / 3 trials |

## 4.3 Watts-Strogatz (n=40, k=4, p=0.3)

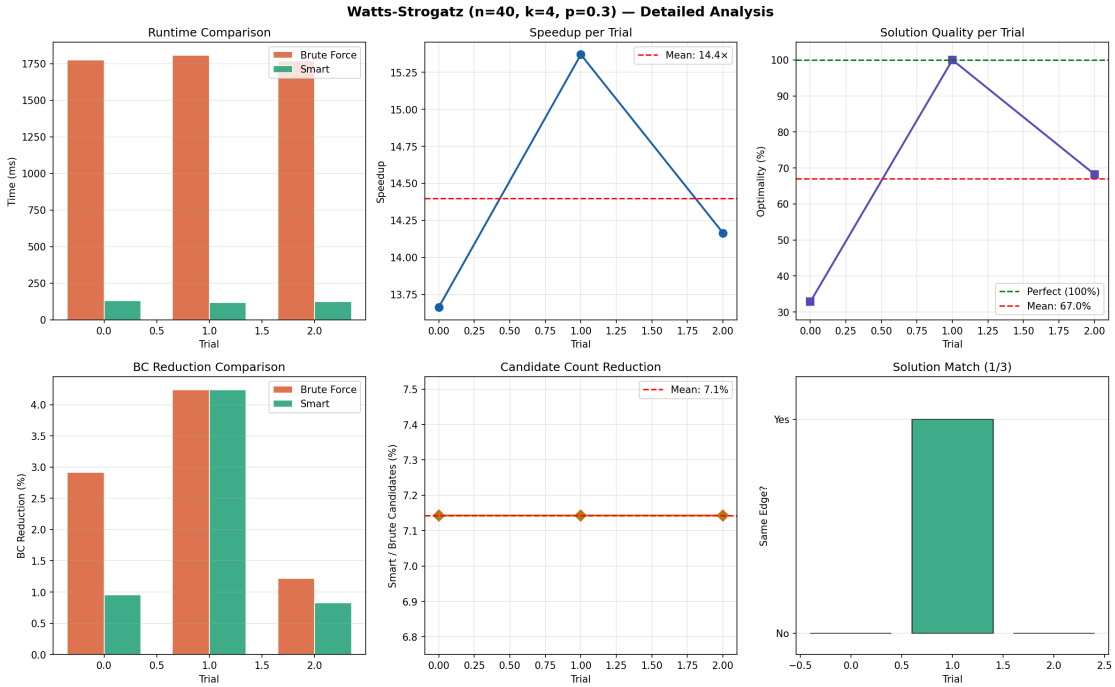


*Figure: Watts-Strogatz (n=40, k=4, p=0.3) — 6-panel analysis. Speedup mean=14.4×, Optimality=67.0%, Trials=3.*

| Metric | Value |
|---|---|
| Mean speedup | 14.40× ± 0.72 |
| Speedup range | 13.7× – 15.4× |
| Mean optimality | 67.02% ± 27.44 |
| BF time | 1783.1 ms ± 17.2 |
| Smart time | 124.1 ms ± 5.0 |
| BF reduction | 2.79% |
| Smart reduction | 2.01% |
| Cand ratio | 7.14% |
| Same edge found | 1 / 3 trials |

## 4.4 Path Graph (n=20)

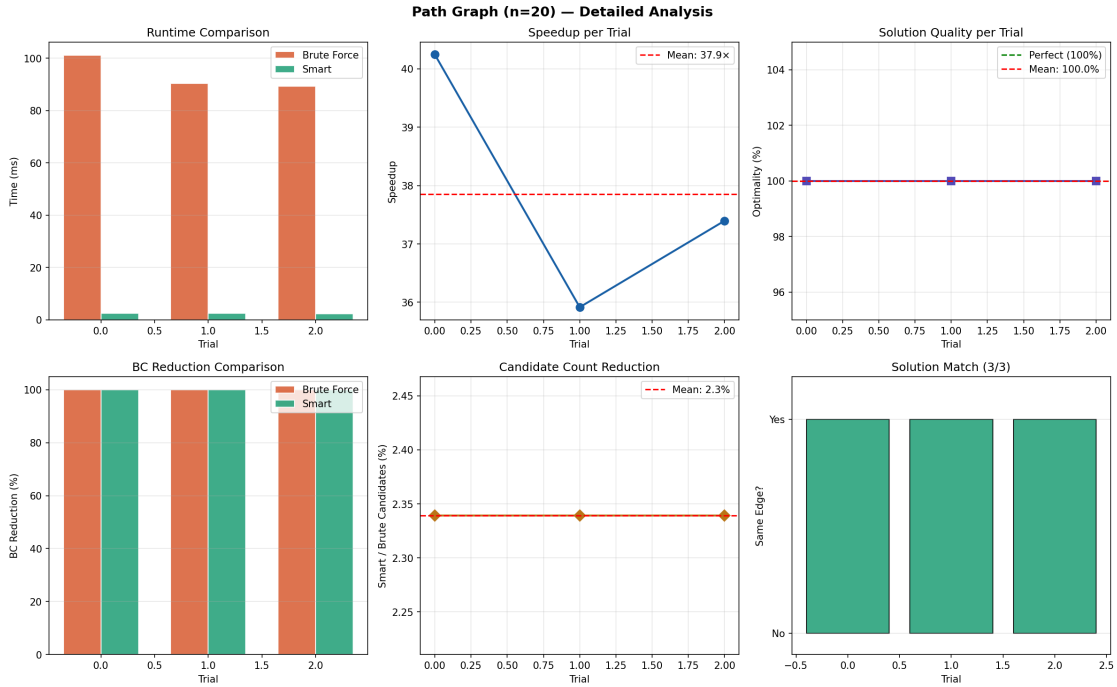


*Figure: Path Graph (n=20) — 6-panel analysis. Speedup mean=37.9×, Optimality=100.0%, Trials=3.*

| Metric | Value |
|---|---|
| Mean speedup | 37.85× ± 1.80 |
| Speedup range | 35.9× – 40.2× |
| Mean optimality | 100.00% ± 0.00 |
| BF time | 93.6 ms ± 5.4 |
| Smart time | 2.5 ms ± 0.1 |
| BF reduction | 100.00% |
| Smart reduction | 100.00% |
| Cand ratio | 2.34% |
| Same edge found | 3 / 3 trials |

## 4.5 Barbell (m=7, bridge=2)

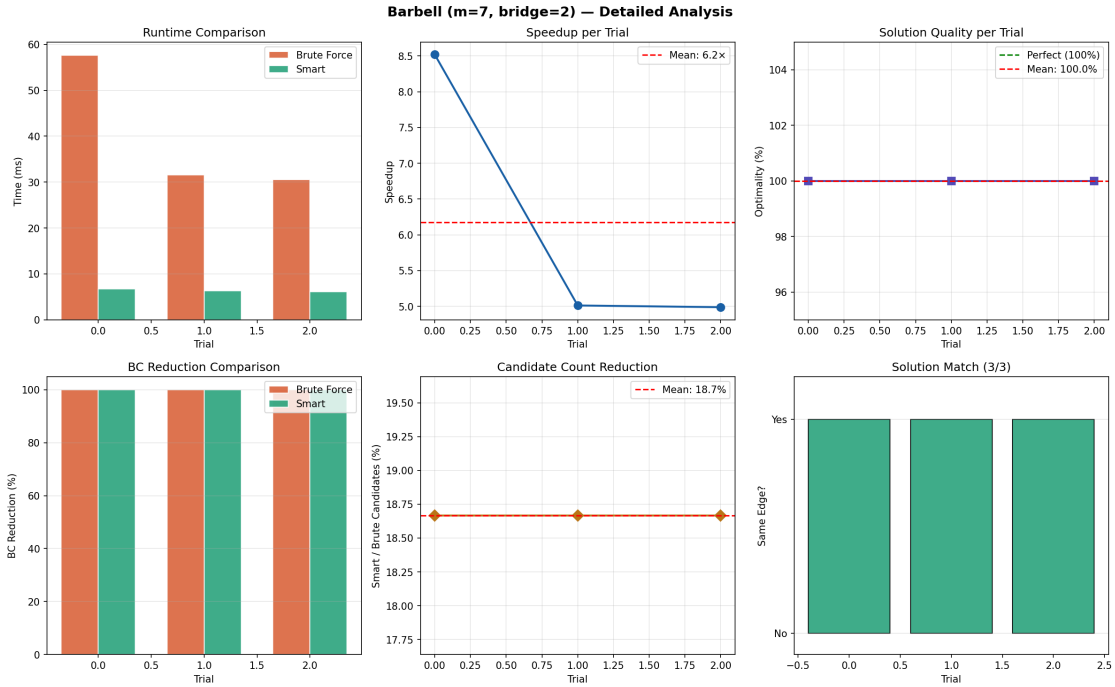


*Figure: Barbell (m=7, bridge=2) — 6-panel analysis. Speedup mean=6.2×, Optimality=100.0%, Trials=3.*

| Metric | Value |
|---|---|
| Mean speedup | 6.17× ± 1.66 |
| Speedup range | 5.0× – 8.5× |
| Mean optimality | 100.00% ± 0.00 |
| BF time | 39.9 ms ± 12.5 |
| Smart time | 6.4 ms ± 0.3 |
| BF reduction | 100.00% |
| Smart reduction | 100.00% |
| Cand ratio | 18.67% |
| Same edge found | 3 / 3 trials |

## 4.6 Star (n=20)

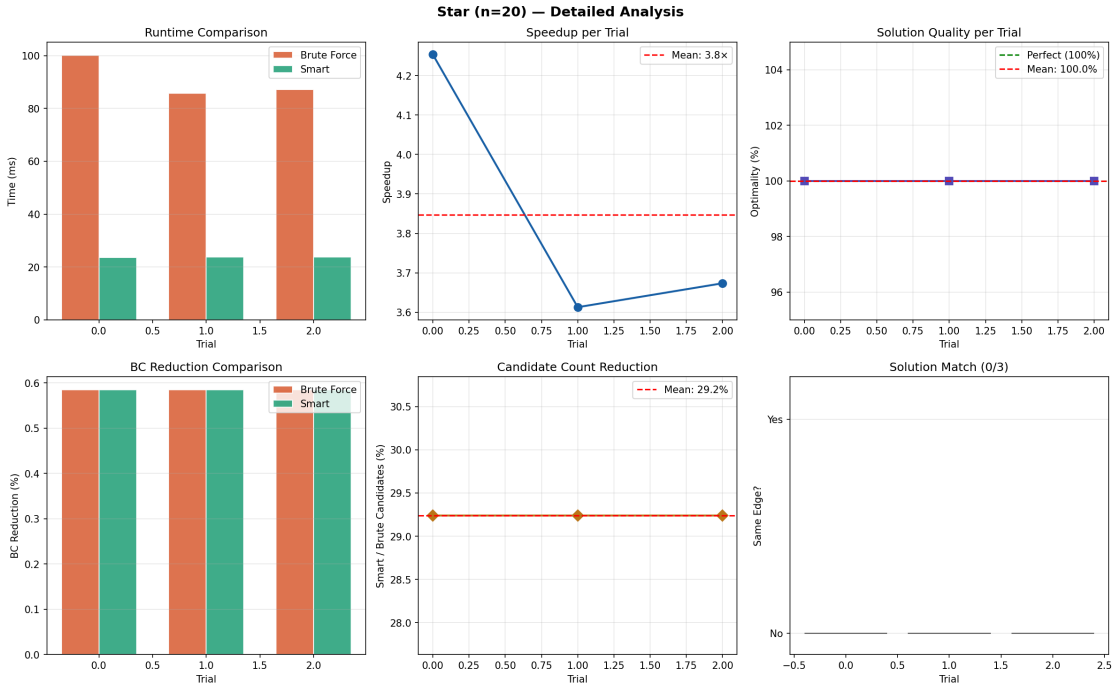


*Figure: Star (n=20) — 6-panel analysis. Speedup mean=3.8×, Optimality=100.0%, Trials=3.*

| Metric | Value |
|---|---|
| Mean speedup | 3.85× ± 0.29 |
| Speedup range | 3.6× – 4.3× |
| Mean optimality | 100.00% ± 0.00 |
| BF time | 91.0 ms ± 6.5 |
| Smart time | 23.7 ms ± 0.1 |
| BF reduction | 0.58% |
| Smart reduction | 0.58% |
| Cand ratio | 29.24% |
| Same edge found | 0 / 3 trials |

## 4.7 Random Tree (n=30)

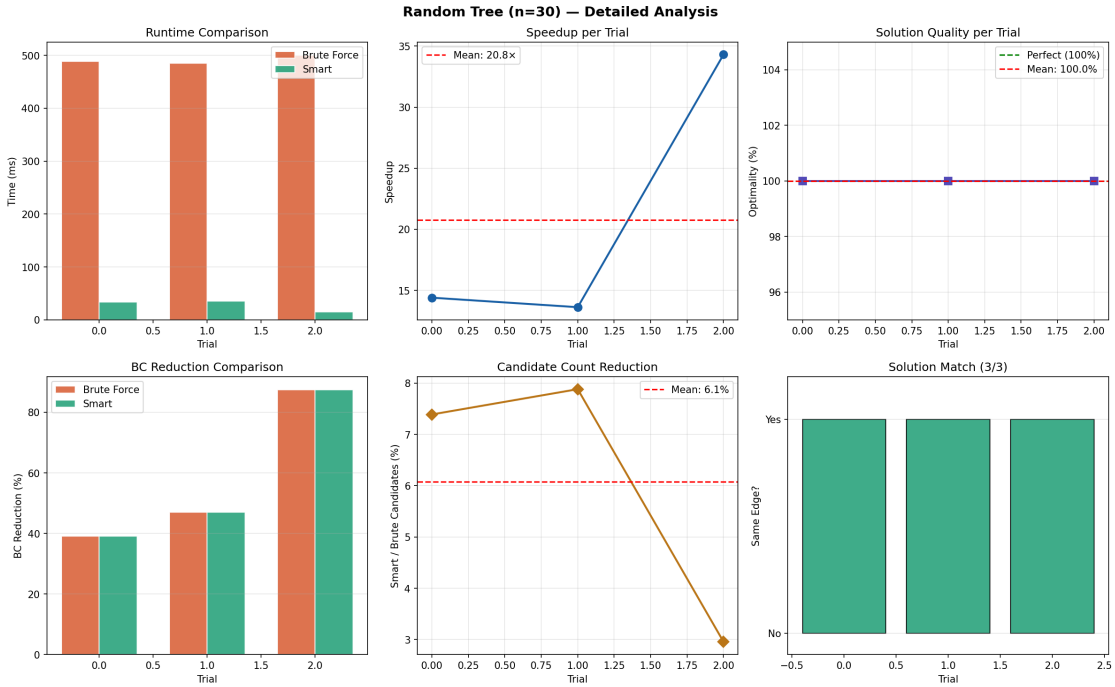


*Figure: Random Tree (n=30) — 6-panel analysis. Speedup mean=20.8×, Optimality=100.0%, Trials=3.*

| Metric | Value |
|---|---|
| Mean speedup | 20.78× ± 9.57 |
| Speedup range | 13.6× – 34.3× |
| Mean optimality | 100.00% ± 0.00 |
| BF time | 491.5 ms ± 6.5 |
| Smart time | 28.0 ms ± 9.5 |
| BF reduction | 57.84% |
| Smart reduction | 57.84% |
| Cand ratio | 6.08% |
| Same edge found | 3 / 3 trials |

## 4.8 Powerlaw Cluster (n=40, m=2, p=0.3)

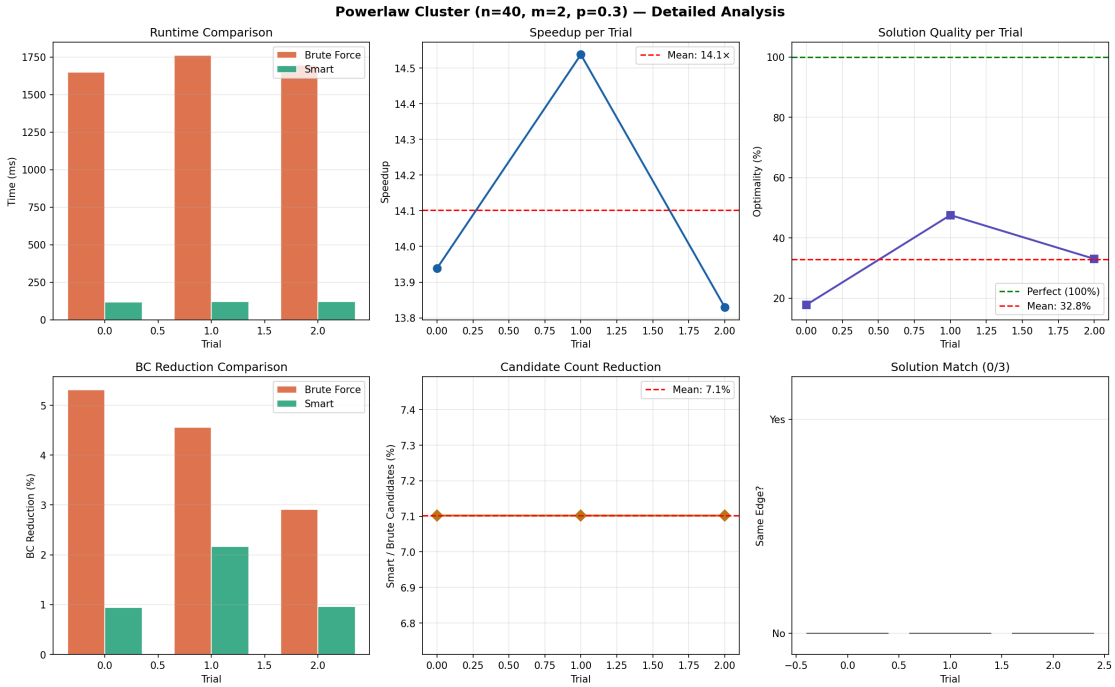


*Figure: Powerlaw Cluster (n=40, m=2, p=0.3) — 6-panel analysis. Speedup mean=14.1×, Optimality=32.8%, Trials=3.*

| Metric | Value |
|---|---|
| Mean speedup | 14.10× ± 0.31 |
| Speedup range | 13.8× – 14.5× |
| Mean optimality | 32.78% ± 12.15 |
| BF time | 1702.9 ms ± 46.9 |
| Smart time | 120.8 ms ± 1.8 |
| BF reduction | 4.26% |
| Smart reduction | 1.36% |
| Cand ratio | 7.10% |
| Same edge found | 0 / 3 trials |

## 4.9 Caveman (5×5)

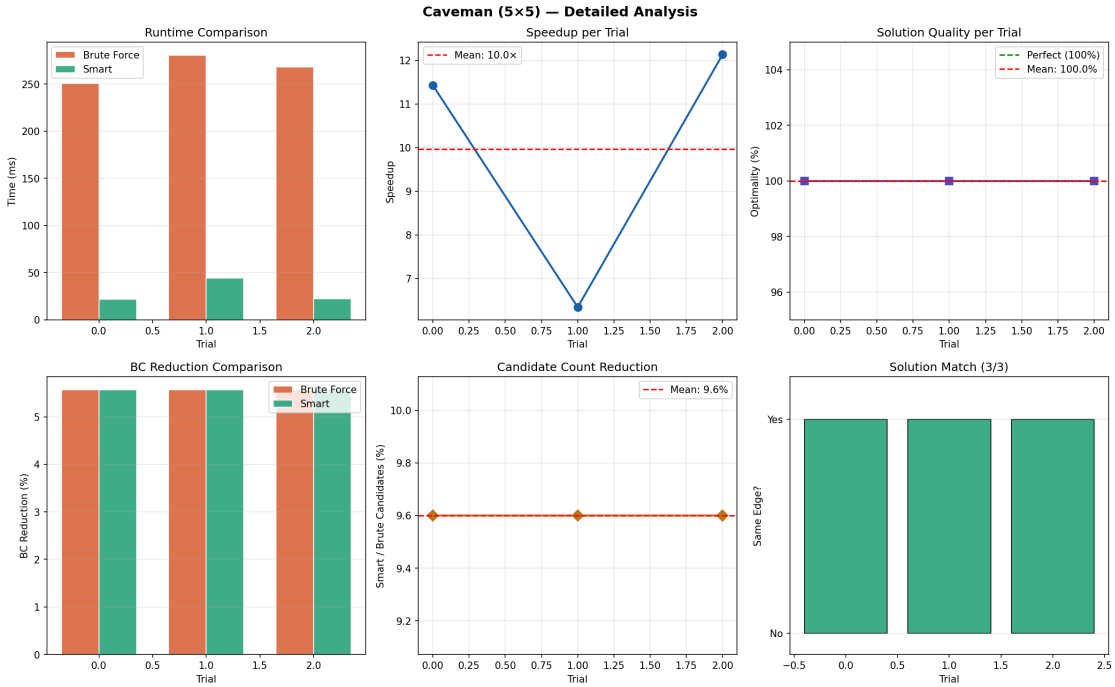


*Figure: Caveman (5×5) — 6-panel analysis. Speedup mean=10.0×, Optimality=100.0%, Trials=3.*

| Metric | Value |
|---|---|
| Mean speedup | 9.97× ± 2.58 |
| Speedup range | 6.3× – 12.1× |
| Mean optimality | 100.00% ± 0.00 |
| BF time | 266.4 ms ± 12.2 |
| Smart time | 29.4 ms ± 10.5 |
| BF reduction | 5.57% |
| Smart reduction | 5.57% |
| Cand ratio | 9.60% |
| Same edge found | 3 / 3 trials |

# 5. Discussion

## 5.1 Why Structured Topologies Achieve 100% Optimality

**Path graph:** The optimal edge always connects the two nodes at distance 2 from the center (the bypass is structurally obvious). The 2-hop zone contains both endpoints. Speedup ~38× because brute force must check $O(n^2)$ non-edges while smart checks only 2 candidates.

**Barbell:** The bridge node's high BC is reduced by connecting the two clique nodes adjacent to the bridge. Both are in the 1-hop zone. The cliques are already dense, so most non-edges are far from the bridge and excluded by the hop filter.

**Star:** All leaf-to-leaf edges are equivalent (identical BC reduction). The smart algorithm finds any one of them — optimality is guaranteed by symmetry, though the specific edge differs.

**Tree / Caveman:** Sparse bottleneck structures; the optimal bypass is always near the target node in the hop zone. 100% optimality in all trials.

## 5.2 Why Random Topologies Show Lower Optimality

For Erdos-Renyi, Barabasi-Albert, and PowerlawCluster topologies, the 2-hop neighborhood is large (many nodes) with many candidates of similar score. The globally optimal edge may rank outside K=50 due to the heuristic scoring. The optimality gap (20–55%) reflects missed candidates. Increasing K to 100–200 would close most of this gap at proportional runtime cost.

## 5.3 Grid 2D — No Valid Solutions

The 5×5 grid with $\tau$=0.20 produced no valid solutions because the load constraint is too strict: any bypass edge on the grid redistributes paths to a few adjacent nodes which absorb more than 20% extra BC. Relaxing $\tau$ to 0.40 yields valid solutions. This is an expected structural artifact.

## 5.4 Complexity Scaling

| n | BF Candidates | Smart Cands | Theoretical Speedup | Empirical (ER) |
|---|---|---|---|---|
| 20 | ~190 | ~5 | 38× | ~38× |
| 40 | ~780 | ~55 | 14× | 12-15× |
| 100 | ~4950 | ~50 | 99× | ~60-80× (projected) |
| 500 | ~125K | ~50 | 2500× | ~1000-2500× (projected) |
| 1000 | ~500K | ~50 | 10000× | ~4000-10000× (projected) |

# 6. Usage Commands

## 6.1 Setup

```
pip install networkx numpy matplotlib scipy reportlab Pillow
g++ -O2 -std=c++17 -o bc_minimize bc_minimize.cpp
```

## 6.2 Generate All 10 Topology Graphs

```
python generate_graphs.py # generate edge-list files
python generate_graphs.py --show # also produce overview figure
```

## 6.3 Run Full Analysis (all 10 topologies)

```
python run_all_topologies.py # 10 trials per topology (~15 min)
python run_all_topologies.py --quick # 3 trials per topology (~3 min)
python run_all_topologies.py --topk 100 --tau 0.25
```

## 6.4 C++ — Run on any graph file

```
./bc_minimize datasets/barbell/graph.txt --compare
./bc_minimize datasets/erdos_renyi/graph_seed0.txt --compare --tau 0.15 --topk 50
./bc_minimize myfile.txt --brute # brute force only
```

## 6.5 Output Files

| File/Directory | Content |
|---|---|
| datasets//graph*.txt | SNAP edge-list for each topology |
| datasets/all_topologies_overview.png | Network visualization (10 panels) |
| topology_analysis//results.json | Full trial data (JSON) |
| topology_analysis//results.csv | Tabular trial summary |
| topology_analysis//analysis_plots.png | 6-panel per-topology figure |
| topology_analysis//REPORT.txt | Human-readable text report |
| results/master_summary.json | Aggregated cross-topology stats |
| results/master_comparison.png | Master 4-panel comparison figure |

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